Agilex® 5 FPGAs and SoCs E-Series Product Table

Version 2026.08.07

Product Line		Device Group A FPGAs				
		A5E 013A	A5E 028A	A5E 043A	A5E 052A	A5E 065A
Resources	Specification Options[1]	C, D, I	D, I	C, D, K, L	C, D, K, L	C, D, K, L
	Logic elements (LEs)	138,060	282,256	434,240	523,920	656,080
	Adaptive logic modules (ALMs)	46,800	95,680	147,200	177,600	222,400
	ALM registers	187,200	382,720	588,800	710,400	889,600
	M20K memory blocks	358	716	1,050	1,288	1,611
	M20K memory size (Mb)	6.99	13.98	20.51	25.16	31.46
	MLAB memory count	2,340	4,784	6,720	8,440	11,120
	MLAB memory size (Mb)	1.43	2.92	4.10	5.15	6.79
	I/O PLL	4	4	8	8	8
	Fabric-feeding I/O PLL[2]	8	10	13	13	13
	Variable-precision digital signal processing (DSP) blocks	188	376	564	676	846
	18 x 19 multipliers	376	752	1,128	1,352	1,692
	Peak INT8 (TOPS)	5.78	11.55	17.33	20.78	25.99
Maximum Available Device Resources	LVDS pairs at 1.6 Gbps	96	96	192	192	192
	DDR4/5 and LPDDR4/5 interfaces (x32)	2	2	4	4	4
	MIPI D-PHY interface	14	14	28	28	28
	Differential (RX or TX) pairs at 28 Gbps	4	12	16	24	24
	PCIe 4.0 x4 instance	1	3	4	6	6
	High-speed I/O (HSIO)	192	192	384	384	384
	High-voltage I/O (HVIO)	200	200	120	120	120
	Secure device manager (SDM)	Provides SHA-384 bitstream integrity, ECDSA 256/384 bitstream authentication, AES-256 bitstream encryption, physically unclonable function (PUF) protected key storage, side-channel attack resistance, SPDM attestation, cryptographic services, physical anti-tamper support				
	Hard processor system (HPS)	Multi-core with 32-bit/64-bit dual-core Arm Cortex*-A55 up to 1.5 GHz with 32 KB I/D cache and 128 KB L2 cache, and dual-core Arm Cortex-A76 up to 1.8 GHz with 64 KB I/D cache and 256 KB L2 cache, and up to 2 MB L3 shared cache, multi-channels direct memory access (DMA), 512 KB on-chip RAM, USB 3.1 x1, USB 2.0 OTG x2, TSN MAC x3, UART x2, SPI M x2, SPI S x2, I3C x2, I2C x5, NAND x1, SDMMC x1, Osc timer x2, SP timer x2, watchdog x5, GPIO x2.				
	Transceiver	PCI Express* (PCIe*) hard IP up to PCIe 4.0 x4 EP and RP Transceiver channel count: up to 24 channels at 28 Gbps (NRZ) Ethernet IP: up to 6 x10/25 GbE hard IP (MAC, PCS, and FEC)				

Notes:

1. Specification code definition:

Code	HPS	Transceiver	Cryptographic Services	PQC (013)	PQC (028)	PQC (043/052/065)
C	No	Yes	No	No	–	No
D	Quad HPS	Yes	Yes	Yes	Yes	No
I	No	Yes	No	Yes	Yes	–
K	No	Yes	No	–	–	Yes
L	Quad HPS	Yes	Yes	–	–	Yes

Cryptographic Services: User mode access to SDM hardened cryptographic IP
Quad HPS: 2x Arm Cortex-A55 and 2x Arm Cortex-A76
PQC: Post Quantum Cryptography secure boot
013: -CS suffix OPNs do not support PQC

2. The fabric-feeding IOPLL count inclusive of system PLL at transceiver bank, the System PLL can be repurposed for core fabric usage if not used for transceiver.

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Product Line	Device Group A FPGAs				
	A5E 013A	A5E 028A	A5E 043A	A5E 052A	A5E 065A
Package Options[3] and I/O Pins					
Package code (Package size, ball pitch)	**HVIO/ HSIO[4]/HPSIO/Transceivers**				
B23A (23 mm x 23 mm, Variable[5, 6])	120/96/48/4	120/96/48/12	120/96/48/12	120/96/48/12	120/96/48/12
B32A (32 mm x 32 mm, Variable[5, 6])	200/192/48/4	200/192/48/12	120/384/48/16	120/384/48/24	120/384/48/24

Notes:

3. For more information about the device migration path, please refer to the Device Migration Guidelines: Agilex® 5 FPGAs and SoCs E-Series.
4. The number of LVDS pairs is half of the HSIO count.
5. VPBGA packaging is compatible with Type III PCBs, that use design rules equivalent to 0.8mm ball pitch packages and standard plated through hole (PTH) vias.
6. The ball pitch is variable, ranging from 0.65mm – 1.45mm in single package to ease signal routing. For more details about Variable Pitch BGA (VPBGA) technology, please refer to PCB Design Guidelines (HSSI, EMIF, MIPI, True Differential, PDN) User Guide: Agilex® 5 FPGAs and SoCs.
7. All data is correct at the time of printing, and may be subject to change without prior notice. For the latest information, please visit www.altera.com/fpga.

Agilex® 5 FPGAs and SoCs E-Series Product Table

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Product Line		Device Group B FPGAs							
		A5E 005B	A5E 007B	A5E 008B	A5E 013B	A5E 028B	A5E 043B	A5E 052B	A5E 065B
Resources	Specification Options[1]	A, G	A, G	A, B, C, D, E, H, I	A, B, C, D, E, H, I	B, D, E, H, I	C, D, K, L	C, D, K, L	C, D, K, L
	Logic elements (LEs)	50,445	69,030	85,196	138,060	282,256	434,240	523,920	656,080
	Adaptive logic modules (ALMs)	17,100	23,400	28,880	46,800	95,680	147,200	177,600	222,400
	ALM registers	68,400	93,600	115,520	187,200	382,720	588,800	710,400	889,600
	M20K memory blocks	130	179	229	358	716	1,050	1,288	1,611
	M20K memory size (Mb)	2.54	3.50	4.47	6.99	13.98	20.51	25.16	31.46
	MLAB memory count	850	1,170	1,780	2,340	4,784	6,720	8,440	11,120
	MLAB memory size (Mb)	0.52	0.71	1.09	1.43	2.92	4.10	5.13	6.79
	I/O PLL	2	2	4	4	4	8	8	8
	Fabric-feeding I/O PLL[2]	5	5	8	8	10	13	13	13
	Variable-precision digital signal processing (DSP) blocks	65	94	116	188	376	564	676	846
	18 x 19 multipliers	130	188	232	376	752	1,128	1,352	1,692
	Peak INT8 (TOPS)	1.7	2.46	3.05	4.93	9.85	14.78	17.72	22.17
Maximum Available Device Resources	LVDS pairs at 1,250 Mbps	48	48	96	96	96	192	192	192
	DDR4 and LPDDR4/5 interfaces (x32)	1	1	2	2	2	4	4	4
	MIPI D-PHY interface	7	7	14	14	14	28	28	28
	Differential (RX or TX) pairs at 17 Gbps	0	0	4	4	12	16	24	24
	PCIe 4.0 x4 instance	0	0	1	1	3	4	6	6
	High-speed I/O (HSIO)	96	96	192	192	192	384	384	384
	High-voltage I/O (HVIO)	160	160	200	200	200	120	120	120
	Secure device manager (SDM)	Provides SHA-384 bitstream integrity, ECDSA 256/384 bitstream authentication, AES-256 bitstream encryption, physically unclonable function (PUF) protected key storage, side-channel attack resistance, SPDM attestation, cryptographic services, physical anti-tamper support							
	Hard processor system (HPS)	NA		Multi-core with 32-bit/64-bit dual-core Arm Cortex-A55 up to 1.25 GHz with 32 KB I/D cache and 128 KB L2 cache, and dual-core Arm Cortex-A76 up to 1.4 GHz with 64 KB I/D cache and 256 KB L2 cache, and up to 2 MB L3 shared cache, multi-channels direct memory access (DMA), 512 KB on-chip RAM, USB 3.1 x1, USB 2.0 OTG x2, TSN MAC x3, UART x2, SPI M x2, SPI S x2, I3C x2, I2C x5, NAND x1, SDMMC x1, Osc timer x2, SP timer x2, watchdog x5, GPIO x2.					
	Transceiver	NA		PCIe hard IP up to PCIe 4.0 x4 EP and RP Transceiver channel count: up to 24 channels at 17 Gbps (NRZ) Ethernet IP: up to 6 x10 GbE hard IP (MAC, PCS, and FEC)					

Notes:

1. Specification code definition:

Code	HPS	Transceiver	Cryptographic Services	PQC (008/013)	PQC (028)	PQC (043/052/065)
A	No	No	No	No	–	–
B	Quad HPS	No	Yes	Yes	Yes	–
C	No	Yes	No	No	–	No
D	Quad HPS	Yes	Yes	Yes	Yes	No
E	Dual HPS	No	Yes	Yes	Yes	–
G	No	No	Yes	–	–	–
H	No	No	No	Yes	Yes	–
I	No	Yes	No	Yes	Yes	–
K	No	Yes	No	–	–	Yes
L	Quad HPS	Yes	Yes	–	–	Yes

Cryptographic Services: User mode access to SDM hardened cryptographic IP
Quad HPS: 2x Arm Cortex-A55 and 2x Arm Cortex-A76
Dual HPS: 2x Arm Cortex-A55
Code G: Only for package B18A
PQC: Post Quantum Cryptography secure boot
008/013: -CS suffix OPNs do not support PQC

2. The fabric-feeding IOPLL count inclusive of system PLL at transceiver bank, the System PLL can be repurposed for core fabric usage if not used for transceiver.

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Product Line	Device Group B FPGAs							
	A5E 005B	A5E 007B	A5E 008B	A5E 013B	A5E 028B	A5E 043B	A5E 052B	A5E 065B
Package Options[3] and I/O Pins								
Package code (Package size, ball pitch)	**HVIO/ HSIO[4]/HPSIO/Transceivers**							
B15A (15 mm x 15 mm, Variable[5, 6])	80/72/0/0	80/72/0/0						
M16A (16 mm x 16 mm, 0.5 mm[7])			40/192/48/4	40/192/48/4	40/192/48/ 8			
B18A (18mm x 18mm, Variable[5, 6])	160/48/0/0	160/48/0/0						
B23B (23 mm x 23 mm, Variable[5, 6])	159/96/0/0	159/96/0/0	159/192/48/0	159/192/48/0	159/192/48/0			
B23A (23 mm x 23 mm, Variable[5, 6])			120/96/48/4	120/96/48/4	120/96/48/12	120/96/48/12	120/96/48/12	120/96/48/12
B32A (32 mm x 32 mm, Variable[5, 6])			200/192/48/4	200/192/48/4	200/192/48/12	120/384/48/16	120/384/48/24	120/384/48/24

Notes:

3. For more information about the device migration path, please refer to the Device Migration Guidelines: Agilex® 5 FPGAs and SoCs E-Series.
4. The number of LVDS pairs is half of the HSIO count.
5. VPBGA packaging is compatible with Type III PCBs, that use design rules equivalent to 0.8mm ball pitch packages and standard plated through hole (PTH) vias.
6. The ball pitch is variable, ranging from 0.65mm – 1.45mm in single package to ease signal routing. For more details about Variable Pitch BGA (VPBGA) technology, please refer to PCB Design Guidelines (HSSI, EMIF, MIPI, True Differential, PDN) User Guide: Agilex® 5 FPGAs and SoCs.
7. 0.5mm Micro Fineline BGA (MBGA).
8. All data is correct at the time of printing, and may be subject to change without prior notice. For the latest information, please visit www.altera.com/fpga.

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